12TH CHEMISTRY · CHAPTER 19 · CHAPTER TEST

Basics of Organic Chemistry

Chapter-wise test: 6 MCQs, 5 of 6 short questions, 1 long question

Class: 12th (HSSC Part II)	**Subject:** Chemistry	**Total Marks:** 26
Chapter: 19	**Time Allowed:** 45 minutes	**Name:** _____

OBJECTIVE PART (6 × 1 = 6)

Q.1 Choose the correct answer.

1. The ability of self-linkage of carbon atoms is called:

(a) Isomerism (b) Hybridization
(c) Polymerisation (d) Catenation

2. The IUPAC name of an ester is:

(a) Alkanal (b) Alkyl alkanoate
(c) Alkanone (d) Alkoxyalkane

3. Homolytic fission of a covalent bond produces:

(a) Free radicals (b) Cations
(c) Anions (d) Ion pairs

4. Which of the following is a nucleophile?

(a) Boron trifluoride (b) Proton
(c) Ammonia (d) Sulphur trioxide

5. Which compound shows cis-trans isomerism?

(a) 1-Butene (b) 2-Butene
(c) Propene (d) Ethene

6. A 50 : 50 mixture of two enantiomers is called a:

(a) Meso form (b) Tautomer
(c) Racemic mixture (d) Metamer

SUBJECTIVE PART — SHORT QUESTIONS (5 × 2 = 10)

Q.2 Write short answers to any FIVE (5) questions.

i. Define functional group with one example.
ii. Differentiate between molecular and empirical formula.
iii. Define electrophile and nucleophile.
iv. What is a chiral carbon?
v. Why does but-1-ene not show cis-trans isomerism?
vi. Why is meso tartaric acid optically inactive?

LONG QUESTION (2 × 5 = 10)

Q.3 Answer both parts.

(a) Explain homolytic and heterolytic fission of a covalent bond.
(b) Describe optical isomerism with the example of lactic acid.

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